



ECOFLOW VENTURI DAMPER

ACESIAN TECHNOLOGIES PTE LTD

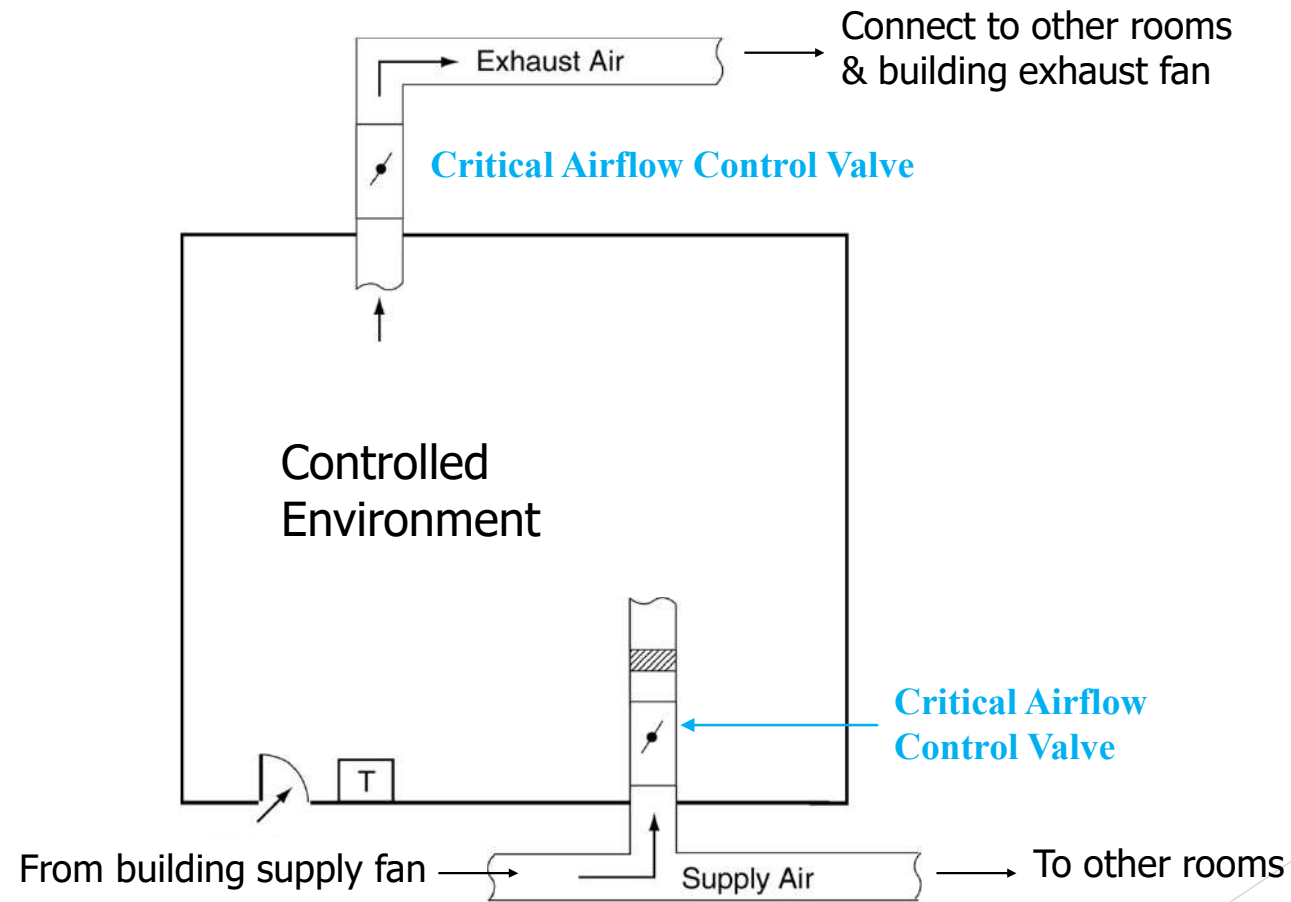
The Ultimate In Critical Airflow Control

CRITICAL ENVIRONMENT

- ▶ Critical Environment, sometimes called a controlled environment, is an area that required precise environmental conditions, such as **Laboratory / Hospital / Pharmaceutical / Life Science / High End Electronic Plant.**
- ▶ Through accurate & precise **Airflow Control** to maintain Room Pressurization, Temperature & Humidity, for the **SAFETY**, as well as asset protection and human comfort



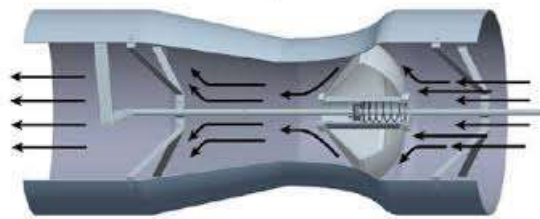
CRITICAL ENVIRONMENT



AIRFLOW CONTROL VALVE



VAV Box with Cross Sensor



Spring & Cone Venturi Valve



**ACESIAN ECOFLOW
VENTURI DAMPER**

CRITICAL AIRFLOW CONTROL VALVE CONSIDERATION

- ▶ Airflow Measurement Accuracy, typically $\pm 5\%$, at Minimum to Maximum Airflow
- ▶ Speed of Response, typically $< 3s$
- ▶ Instant “TRUE” Airflow Measurement, Feedback, Control & Monitor
- ▶ Pressure Independent Control
- ▶ Low Pressure Drop & Good Turndown Ratio
- ▶ Minimum Maintenance & Long-Term Durability
- ▶ Coating for Corrosive Environment Application

Reference: ASHRAE Laboratory Design Guide Second Ed, Section 11: Controls

CHALLENGES WITH VAV BOX

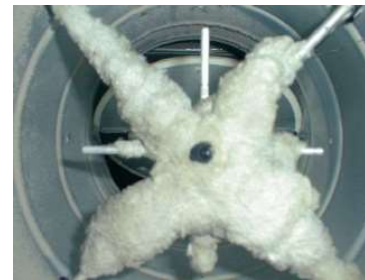
Problems

- ▶ Accuracy at Low Airflow ($< 2.5\text{m/s}$)
- ▶ Sensing element in the air stream
- ▶ Sensor probes require maintenance & cleaning
- ▶ Resistant to contaminants
- ▶ Required min. installation straight length



Risks

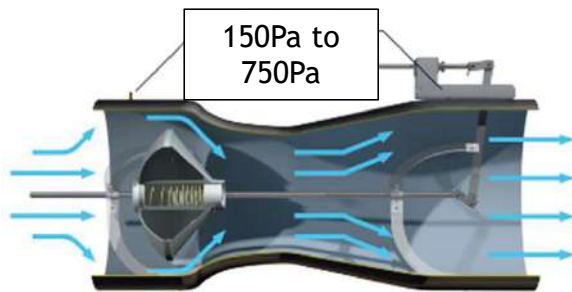
- ▶ Loss of airflow control effectiveness & accuracy
- ▶ Wasted energy
- ▶ Loss of room pressure
- ▶ Loss of research integrity
- ▶ Safety & Health



CHALLENGES WITH VENTURI VALVE

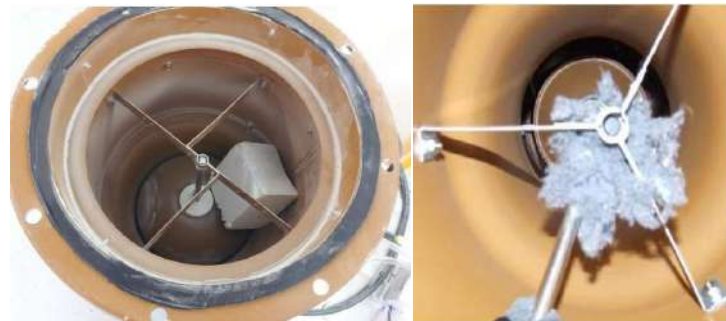
Problems

- ▶ No actual airflow measurement
- ▶ Mechanical parts in the airstream
- ▶ High minimum operating pressure
- ▶ Spring degrade due to aging & fatigue
- ▶ Accumulation of particles on the assembly
- ▶ Installation orientation issues



Risks

- ▶ Wasted energy
- ▶ Loss of airflow control effectiveness & accuracy
- ▶ Loss of mechanical pressure-independent control
- ▶ Higher pressure drop
- ▶ Required maintenance



ECOFLOW VENTURI DAMPER FEATURES

- ▶ **Probe-less, Venturi airflow measurement method, $\pm 5\%$ Accuracy, throughout the entire range from maximum to minimum airflow**
- ▶ **Coupled with high accuracy differential pressure sensor**
- ▶ **Fitted with fast response electric actuator (<3s)**
- ▶ **Pressure Independent Control**
- ▶ **In-House developed Airflow Control through freely programmable controller**



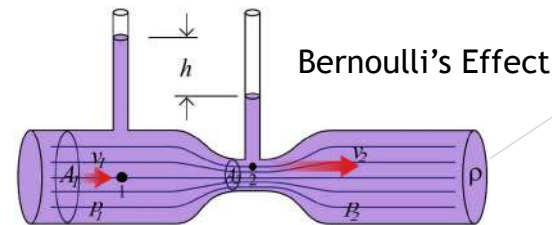
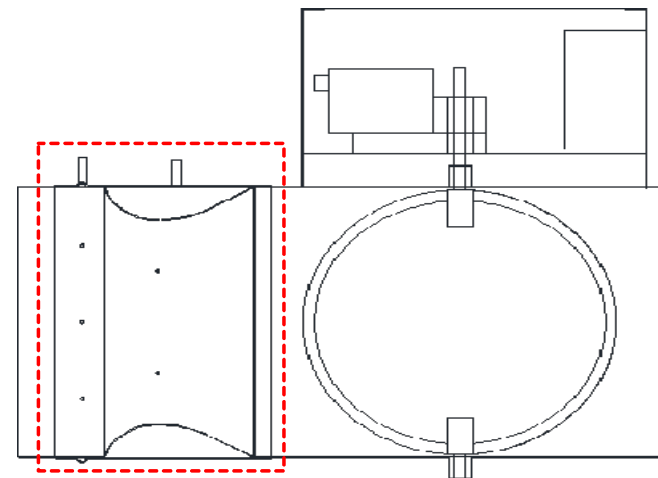
ECOFLOW VENTURI DAMPER FEATURES



- ▶ **Turndown Ratio 10:1**
- ▶ **Minimum Component in the Airstream**
- ▶ **Low Pressure Drop**
- ▶ **Minimum Maintenance**
- ▶ **FM approved ETFE (Ethylene Tetrafluoroethylene) coating**
- ▶ **No Installation Orientation Issue**

VENTURI FLOW MEASUREMENT

- ▶ Venturi = Most common flowmeter
- ▶ The Venturi measures a fluid's flowrate by reducing the cross-sectional flow area in the flow path and generating a pressure difference.
- ▶ The fluid is passed through a pressure recovery exit section where up to 80% of the differential pressure generated at the throat is recovered.
- ▶ Venturi can measure fluid flowrates with a minimal total pressure loss



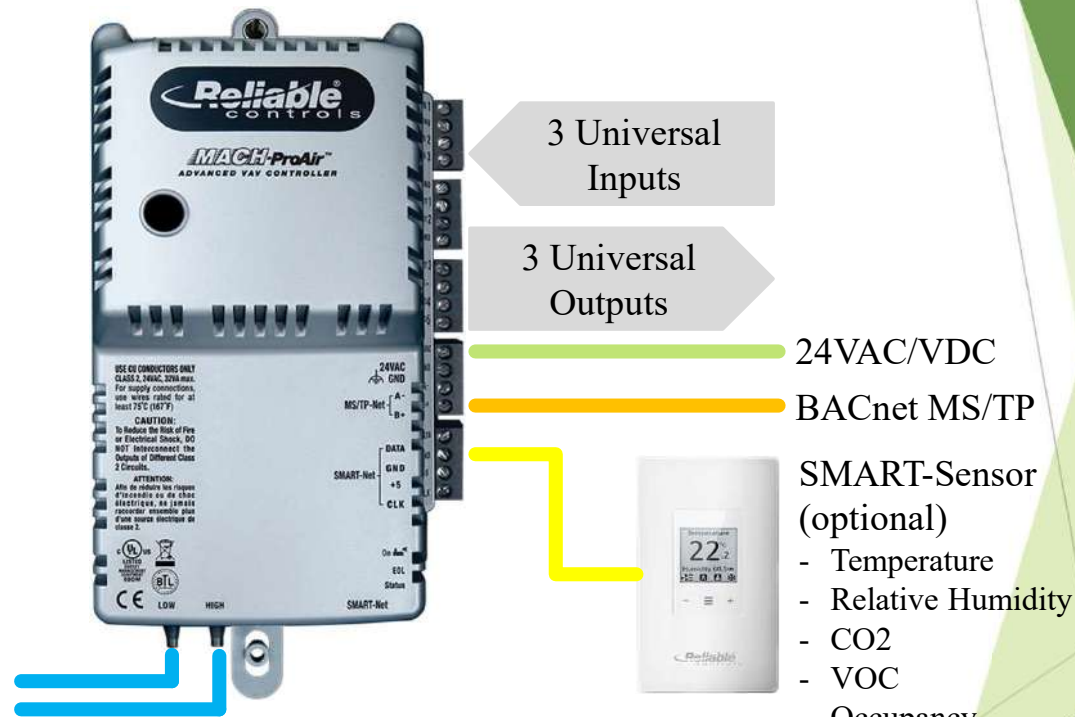
BACnet - BUILDING CONTROLLER (B-BC)

Airflow Control

- ❖ In-House developed airflow control algorithms and/or configurable to meet project specification
- ❖ Configurable P, PI, or PID controls for closed loop control

Pressure Sensor

- ▶ Range: $\pm 0-500$ Pa (0-2" WC)
- ▶ Accuracy: 3% of reading
- ▶ Resolution: 0.122 Pa (0.0005"WC)
- ▶ Maximum zero-point accuracy 0.2 Pa (0.0008"WC)



FAST RESPONSE ELECTRIC ACTUATOR



- ▶ Nominal torque **6Nm** at **2s** running time
- ▶ Operating Voltage: AC/DC 24V
- ▶ Positioning Signal: DC 0(2)-10V, 0(4)...20 mA
- ▶ Positioning Resolution: 250 steps for 90°
- ▶ IP54 Housing
- ▶ Certification: CE, RCM, EN 60730-2-14

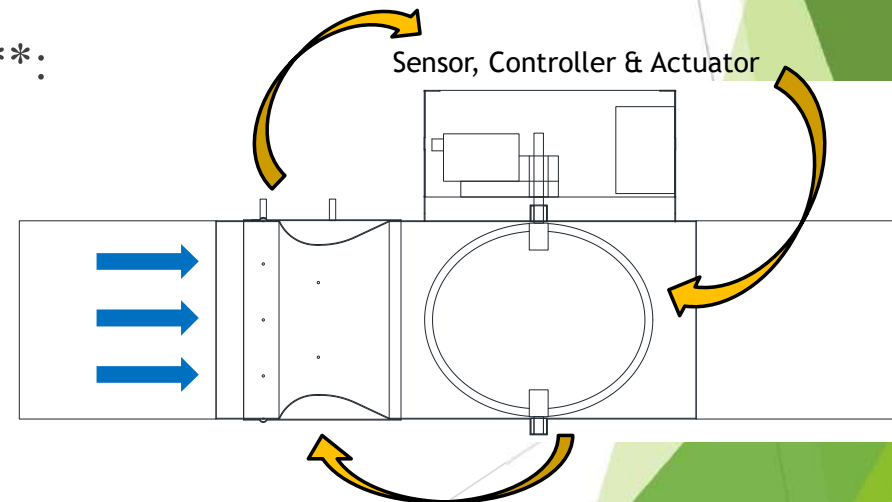
AIRFLOW CONTROL SYSTEM ACCURACY

► Closed Loop Control

- A **closed-loop** or **feedback** control measures actual changes in the controlled variable and actuates the controlled device to bring about a change.*

► Complete Airflow Control System Accuracy**:

- **Sensor;**
- **Controller;**
- **Actuator;**
- **Flow Control Device**



*2021 ASHRAE Handbook – Fundamentals of Control, Chapter 7

** ASHRAE Laboratory Design Guide 2nd Edition; 2016, Chapter 11 - Controls

AIRFLOW CONTROL SYSTEM ACCURACY

- ▶ Overall **airflow measurement accuracy** combines the effects of the **sensing element, pressure sensor & controller signal converter**.
- ▶ **Pressure sensors** have several types of **errors** that should be considered

AIRFLOW CONTROL SYSTEM ACCURACY

► Pressure Sensor Error Example

► 0 - 500Pa (2" wc) with **0.5% Error of Full Scale**

CMH	Actual DP*	Pressure Sensor Error	Measured DP	Measured CMH	% Flow
1800	200.0	2.5	202.5	1811	0.62%
900	50.0	2.5	52.5	922	2.47%
450	12.5	2.5	15.0	493	9.54%
180	2.0	2.5	4.5	270	50.00%

*Differential Pressure in Pa unit

AIRFLOW CONTROL SYSTEM ACCURACY

► Pressure Sensor Error Example

► 0 - 500Pa (2" wc) with **3% Error of Reading**

CMH	Actual DP*	Pressure Sensor Error	Measured DP	Measured CMH	% Flow
1800	200.0	6.0	206.0	1827	1.49%
900	50.0	1.5	51.5	913	1.49%
450	12.5	0.38	12.88	457	1.49%
180	2.0	0.06	2.06	183	1.49%

*Differential Pressure in Pa unit

AIRFLOW CONTROL SYSTEM ACCURACY

- Controller Signal Converter's Resolution: **n-bit**, determine the resolution of Pressure Sensor

n-bit	Converter Resolution	Measurement Voltage Range	Measurement Resolution	Pressure Sensor Range	Pressure Sensor Resolution
8-bit	$2^8 = 256$	0-10VDC	0.039063V	0-500Pa	1.953Pa
10-bit	$2^{10} = 1024$	0-10VDC	0.009765V	0-500Pa	0.488Pa
12-bit	$2^{12} = 4096$	0-10VDC	0.002441V	0-500Pa	0.122Pa
16-bit	$2^{16} = 65536$	0-10VDC	0.000153V	0-500Pa	0.008Pa

AIRFLOW CONTROL SYSTEM ACCURACY

- ▶ Pressure Sensor + Signal Conversion Error Example
 - ▶ **0 - 500Pa (2") Transducer with 3% Error of Reading**
 - ▶ **8-bit Converter**

CMH	Actual DP*	Pressure Sensor Error	Pressure Sensor Resolution	Measured DP	Measured CMH	% Flow
1800	200.000	6.000	1.953	206.000	1827	1.49%
900	50.000	1.500	1.953	51.953	917	1.93%
450	12.500	0.375	1.953	14.453	484	7.53%
180	2.000	0.060	1.953	3.953	253	40.59%

*Differential Pressure in Pa unit

AIRFLOW CONTROL SYSTEM ACCURACY

- ▶ Pressure Sensor + Signal Conversion Error Example
 - ▶ **0 - 500Pa (2")** Transducer with **3% Error of Reading**
 - ▶ **10-bit Converter**

CMH	Actual DP*	Pressure Sensor Error	Pressure Sensor Resolution	Measured DP	Measured CMH	% Flow
1800	200.000	6.000	0.488	206.000	1827	1.49%
900	50.000	1.500	0.488	51.500	913	1.49%
450	12.500	0.375	0.488	12.988	459	1.93%
180	2.000	0.060	0.488	2.488	201	11.53%

*Differential Pressure in Pa unit

AIRFLOW CONTROL SYSTEM ACCURACY

- ▶ Pressure Sensor + Signal Conversion Error Example
 - ▶ **0 - 500Pa (2") Transducer with 3% Error of Reading**
 - ▶ **12-bit Converter**

CMH	Actual DP*	Pressure Sensor Error	Pressure Sensor Resolution	Measured DP	Measured CMH	% Flow
1800	200.000	6.000	0.122	206.000	1827	1.49%
900	50.000	1.500	0.122	51.500	913	1.49%
450	12.500	0.375	0.122	12.875	457	1.49%
180	2.000	0.060	0.122	2.122	185	3.00%

*Differential Pressure in Pa unit

FACTORY ASSEMBLY, TESTING & CALIBRATION

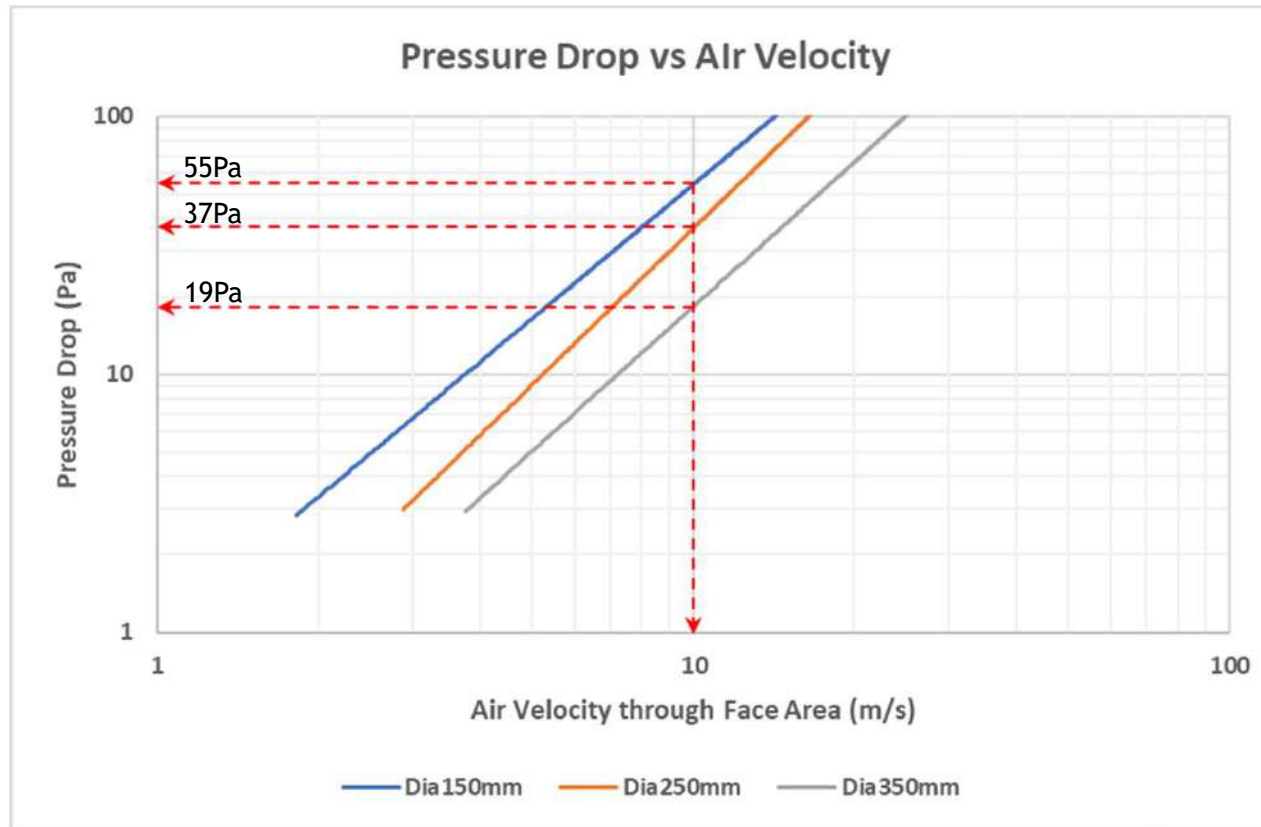
- ▶ ECOFLOW Venturi Damper shall be fully supplied with sensor, controller and actuator, factory assembled / calibrated / tested prior to delivery
- ▶ In-House Test & Calibration Station
 - ▶ High Accuracy ($\pm 0.5\%$) Reference Airflow Station
 - ▶ *N.I.S.T Traceable

*National Institute of Standards and Technology



AIR PERFORMANCE: PRESSURE DROP

- Tested in accordance to AMCA 500-D-07 (Pressure Drop), Figure 5.3



SPEED OF RESPONSE: FUME HOOD CONTROL

ANSI Z9.5 Laboratory Ventilation 2012 Section 6.5.3.2

- ▶ **VAV Response** shall be sufficient to increase or decrease flow within **90%** of the **target flow** or face velocity in a manner that does not increase potential for escape
- ▶ A **response time** of **< 3 sec.** after **completion** of the **sash movement** is considered acceptable for most operations. Faster response times may improve hood containment following the sash movement.



SPEED OF RESPONSE: FUME HOOD CONTROL

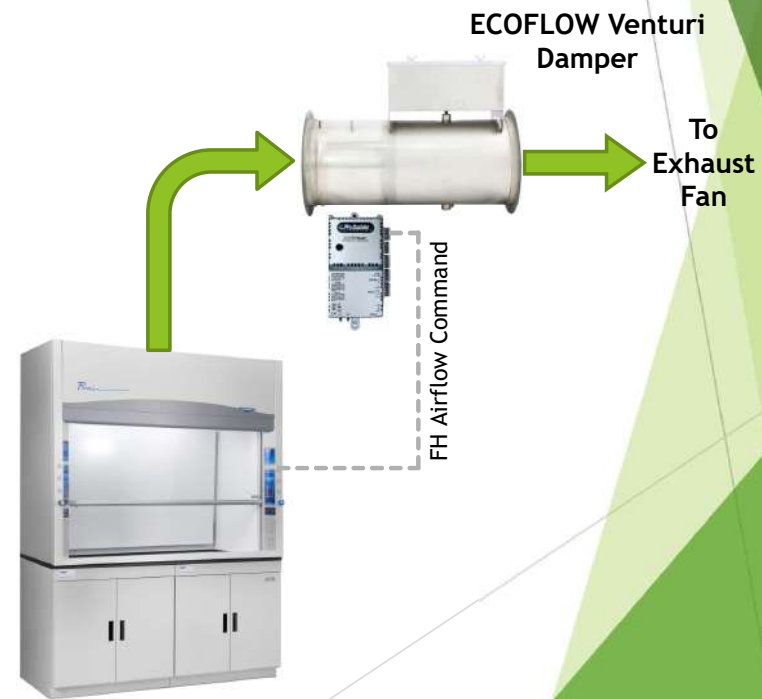
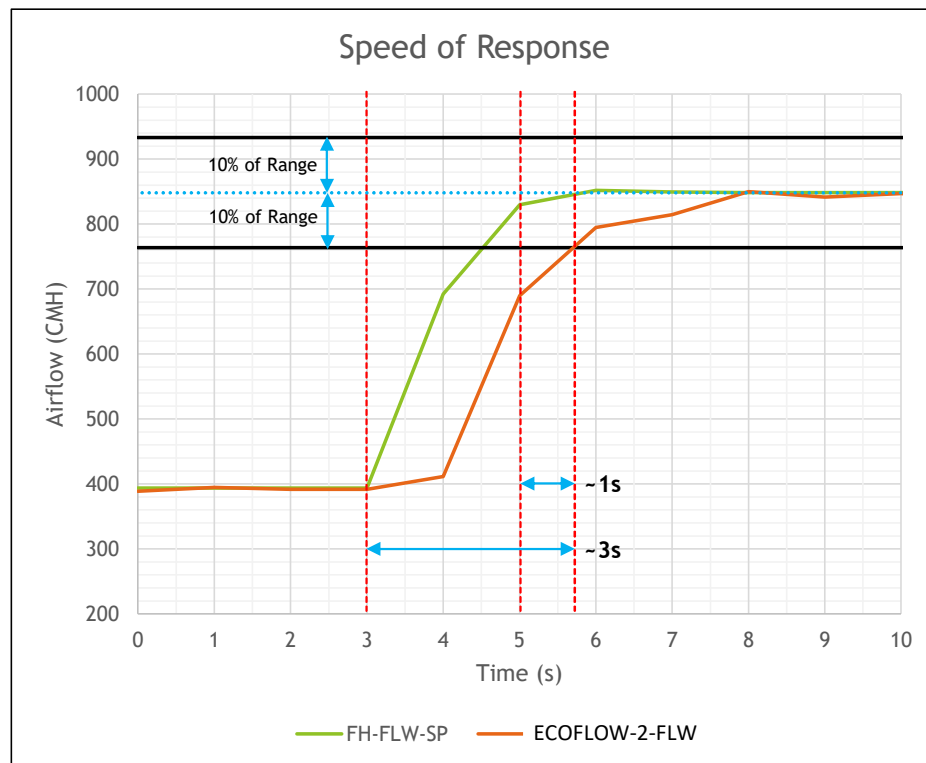
ASHRAE 110-2016 Method of Testing Performance of Laboratory Fume Hood Section 6.3.1

- ▶ **VAV Speed of Response:** the time, measured from the **first movement** of the sash, for the VAV system to restore the slot velocity or airflow to **90%** of the average steady-state value.
- ▶ **VAV Time to Steady State:** the time, measured from the **first movement** of the sash, for the VAV system to restore and maintain the average slot velocity or airflow between **90%** and **110%** of the average steady-state value.



SPEED OF RESPONSE: FUME HOOD CONTROL

- ECOFLOW Venturi Damper able to achieve quick, stable airflow control needed for critical environment



*Data collected from ACESIAN Demo Lab

MAINTENANCE



ECOFLOW Venturi Damper

V.S.

VAV Box with Cross Sensor



Spring & Cone Venturi Valve

OPERATING RANGE

Size	Units	Min	Max	Turndown Ratio
Ø150	CMH	35	342	<u>10:1</u>
Ø200	CMH	87	868	<u>10:1</u>
Ø250	CMH	170	1691	<u>10:1</u>
Ø300	CMH	286	2852	<u>10:1</u>
Ø350	CMH	439	4386	<u>10:1</u>

CONSTRUCTION

- ▶ **Galvanized Steel w/o Angle Flange;**
- ▶ **Stainless Steel 316 c/w Angle Flange;**
- ▶ **Stainless Steel 316 c/w Angle Flange,**
coated with **FM approved ETFE**
(Ethylene Tetrafluoroethylene) coating



SUMMARY

	ECOFLOW	VAV Box	Venturi Valve
Airflow Measurement Accuracy	$\pm 5\%$ (1m/s - 12m/s)	$\pm 5\%$ (2.5m/s - 10m/s)	$\pm 5\%$
Speed of Response (< 3s)	✓	✗	✓
Realtime Measured Airflow Feedback	✓	✓	✗
Pressure Independent Control	✓	✓	✓
Control Turndown	10:1	6:1	16:1
Low Pressure Drop/Operating Pressure	✓	✓	✗

SUMMARY

	ECOFLOW	VAV Box	Venturi Valve
No Sensing Element / Mechanical Parts in Airstream	✓	✗	✗
Minimal Maintenance	✓	✗	✗
Construct in various material	✓	✓	✓
Resistant to Contaminants	✓	✗	✓
No Installation Straight Length Required	✓	✗	✓
No Installation Orientation Issue	✓	✓	✗

PROJECT REFERENCE

SINGAPORE

- NUS AS2
- NUS CeLS
- NUS DSI
- NUS E3A, E7
- NUS EW2
- NUS MD1, MD10, MD11
- NUS MD3, MD4, MD6
- NUS MD9
- NUS S13, S14, S16
- NUS S1A, S2, S3
- NUS S4, S5, S7, S8
- NUS T-Lab
- NUS Wet Science Building
- NUS WS2
- DSO National Laboratories
- Fusionopolis Tower AB
- Jurong Island Laboratories

SINGAPORE

- Westpharma Flurotec
- Westpharma RU Autoclave
- *ThermoFisher Scientific SUT Facility
- SIMTECH in CT2B
- Labcorp Laboratory Fit Out

(*Supply of VAV Controller & Actuator)

TAIWAN

- Animal Health Research Institute
- Chang Gung Medical Center
- Chi Mei Medical Center
- National Cheng Kung University

AUSTRALIA

- University of Sydney



NUS WSB

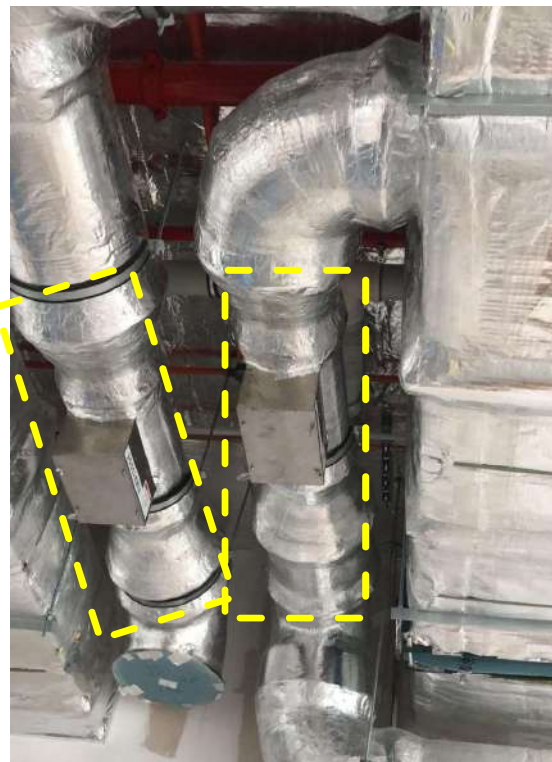


NUS E7

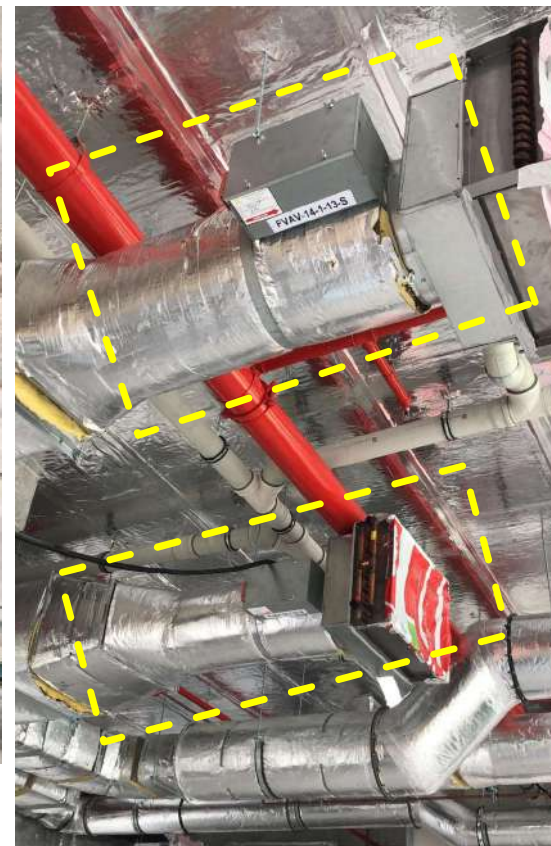
PROJECT REFERENCE



General Exhaust EVD



Fume Hood Exhaust EVD



Supply Air EVD

CONTACT US

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